GROUP 7

PUMA 560 Robot Manipulator

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Project scope

PUMA560: A six revolute joint manipulator with a spherical wrist

- Robot specification and standard DH frames
- Forward and analytical inverse kinematics
- Velocity kinematics and singularity analysis
- Independent checks against MuJoCo

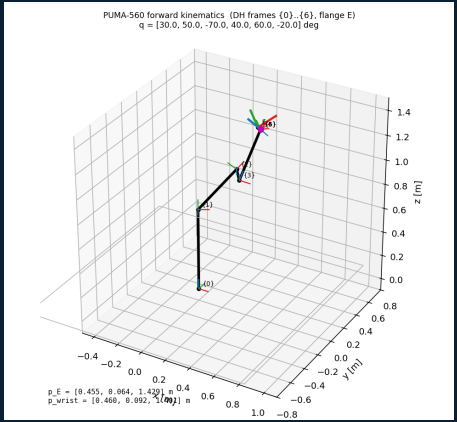
Robot geometry and reference frames

O_0: mounting plane

O_1: $0.67183\,\mathrm{m}$ above O_0

$O_4 = O_5 = O_6$: wrist centre

O_E: $0.04\,\mathrm{m}$ along z_6 from the wrist centre



Standard Denavit–Hartenberg model

$$A_i = R_z(q_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i)$$

Joint	θ	d (m)	a (m)	α (deg)
1	q_1	0.67183	0	90
2	q_2	0	0.4318	0
3	q_3	0.15005	0.0203	-90
4	q_4	0.4318	0	90
5	q_5	0	0	-90
6	q_6	0	0	0

Flange transform: $T_{6E} = \mathrm{Trans}(z, 0.04\,\mathrm{m})$

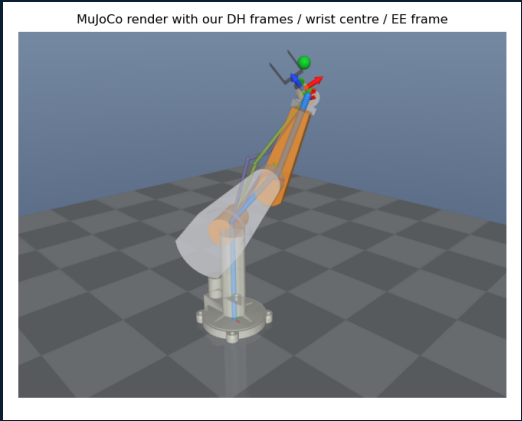
Forward kinematics

$$T_{0E} = A_1 A_2 A_3 A_4 A_5 A_6 T_{6E}$$

Input: six joint angles

Output: end-effector pose, all DH frames and wrist centre

The MuJoCo pose provides an independent comparison



Reachability and joint limits

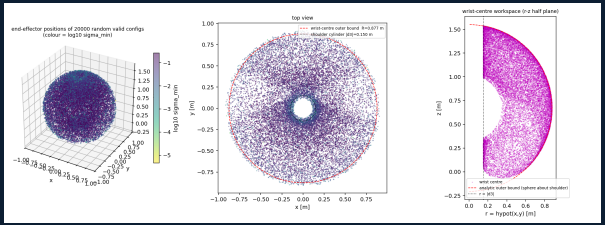
$r \geq |d_3| = 0.15005\,\mathrm{m}$

$|a_2 - L| \leq \rho \leq a_2 + L$

$L = \sqrt{a_3^2 + d_4^2}$

Outer planar reach: $0.8641\,\mathrm{m}$

Joint	Min (°)	Max (°)
J1	-160	160
J2	-45	225
J3	-225	45
J4	-110	170
J5	-100	100
J6	-266	266



A reachable target can still violate every joint-limit branch

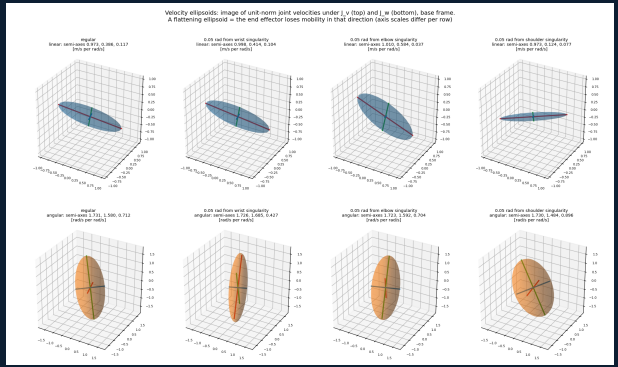
Velocity kinematics

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q)\dot{q}$$

$$J_{v,i} = z_{i-1} \times (p_E - p_{i-1})$$

$$J_{\omega,i} = z_{i-1}$$

Inverse velocity: pseudoinverse or damped least squares



Singularity conditions

$$\det J = a_2 X \left(a_3 \sin q_3 + d_4 \cos q_3\right) \sin q_5$$

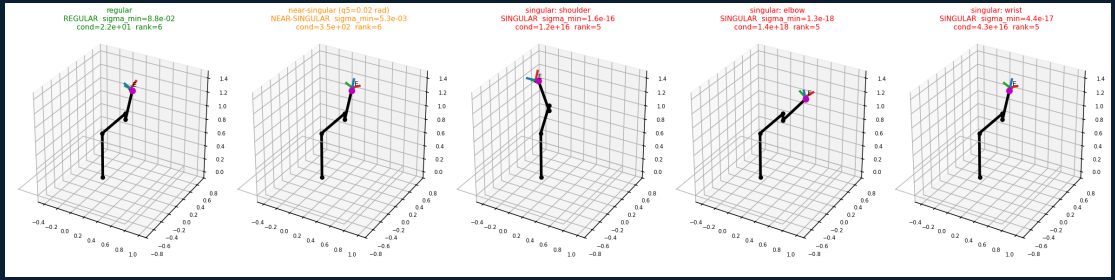
Shoulder: $X = 0$

Wrist centre reaches the cylinder about the waist axis

Elbow: $a_3 \sin q_3 + d_4 \cos q_3 = 0$
Stretched configuration within limits: $q_3 \approx -87.3^\circ$

Wrist: $\sin q_5 = 0$
Joint 4 and joint 6 axes align. Within limits: $q_5 = 0^\circ$

Singular configurations



The Jacobian loses rank and at least one twist direction becomes unavailable

Near-singular joint speeds

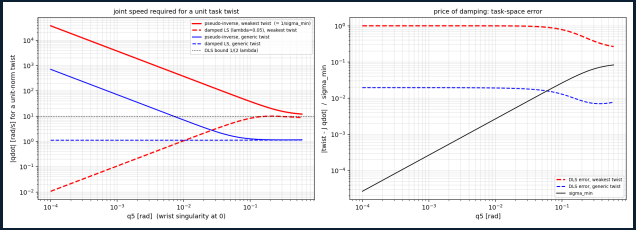
For the weakest twist:

$$\|\dot{q}\| = \frac{\|\text{twist}\|}{\sigma_{\min}}$$

At $q_5 = 0.0001\,\text{rad}$:

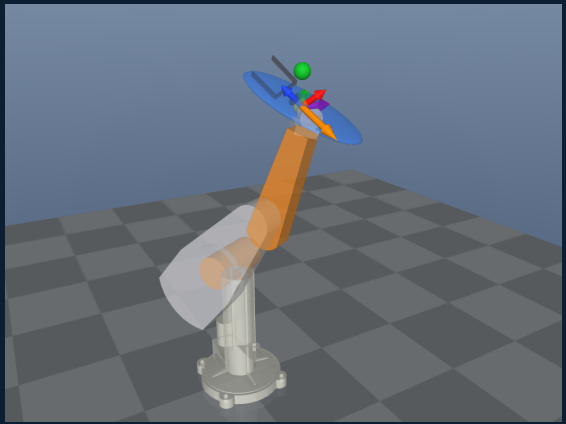
Pseudoinverse: about $37{,}500\,\text{rad/s}$

Damping bounds joint speed at the cost of task error



Interactive MuJoCo demonstration

- Joint-angle or Cartesian input
- DH frames and IK branch overlays
- Velocity ellipsoid and arrows
- Singularity indicator



Numerical verification results

Check	Sample size	Reported result
FK vs MuJoCo	2,000 poses	Max position error $6.8 \times 10^{-16}\,\mathrm{m}$
IK round trip	3,000 poses	3,000 / 3,000 recovered
Jacobian vs differences	1,000 poses	Max absolute error 2.6×10^{-10}
Determinant formula	Report check	Max difference 1.2×10^{-16}

Boundary, unreachable and singular targets also have dedicated checks

Conclusions and limitations

- DH forward kinematics agrees with the MuJoCo reference
- The Jacobian explains lost motion and large joint speeds
- Future work: dynamics, control and obstacle avoidance

References and project sources

Project implementation and verification

`README.md`, `src/`, `run_all.py`
`results/verification_report.md`

Robot parameters

Corke and Armstrong-Helouvry (1994)

Armstrong, Khatib and Burdick (1986)

ARTE and rtb-data: `models/LICENSE_NOTICE.md`